

# Richard Stiskalek<sup>ID</sup>, Curriculum Vitae

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| PROFILE                   | <b>Affiliation:</b> University of Oxford<br><b>NASA ADS:</b> 17 first-author, 16 co-author papers; <i>h</i> -index = 11; citations = 736<br><b>Website:</b> richard-sti.github.io                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Email:</b> richard.stiskalek@physics.ox.ac.uk         |
|                           | <p>I am a Hintze Prize Fellow in Astrophysics (Cosmology) and specialise in Bayesian field-level inference and digital twins of the local Universe to bridge observations and theory. My research combines constrained cosmological simulations, semi-analytic galaxy formation models, simulation-based inference, and machine learning to test galaxy physics, map large-scale flows, and deliver precision measurements of cosmological parameters.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                          |
| EDUCATION                 | <b>University of Oxford</b> (Balliol College), DPhil in Astrophysics<br>Thesis: “Constrained Simulations of the Local Universe as a Laboratory for Precision Cosmology and Astrophysics”<br>Supervised by <i>Harry Desmond</i> , <i>Julien Devriendt</i> , and <i>Adrianne Slyz</i><br><b>Ludwig-Maximilians-Universität München</b> , MSc in Physics<br>Thesis: “Strong-field lensing of gravitational waves” (partially with <i>Albert Einstein Institute Potsdam</i> )<br>Supervised by <i>Miguel Zumalacárregui</i> , <i>Marius A. Oancea</i> , and <i>Jochen Weller</i><br><b>Hong Kong University of Science and Technology</b><br>Undergraduate Exchange Programme in Physics<br><b>University of Glasgow</b> , BSc in Physics with Astrophysics<br>Thesis: “Gravitational-wave cosmology”<br>Supervised by <i>Martin Hendry</i>                                                                                                | 2022 – 2026<br>2020 – 2022<br>2017 – 2018<br>2016 – 2020 |
| ACADEMIC APPOINTMENTS     | <b>University of Oxford</b> , Hintze Centre for Astrophysical Surveys<br>Hintze Prize Fellow in Astrophysics (Cosmology)<br><b>Flatiron Institute</b> , Center for Computational Astrophysics<br>Pre-Doctoral Fellow (with <i>Shy Genel</i> and <i>Lucia A. Perez</i> )<br><b>Institut d’astrophysique de Paris</b><br>Long Term Attachment (with <i>Guilhem Lavaux</i> )<br><b>Max Planck Institute for Gravitational Physics</b> , Observational Relativity and Cosmology<br>Summer Research Intern (with <i>Collin Capano</i> )<br><b>University of Oxford</b> , Sub-department of Astrophysics<br>Summer Research Intern (with <i>Harry Desmond</i> )<br><b>University of Glasgow</b> , Institute for Gravitational Research<br>Summer Research Intern (with <i>John Veitch</i> and <i>Chris Messenger</i> )                                                                                                                       | 2026 – 2029<br>2025<br>2024<br>2020<br>2019<br>2018      |
| COLLABORATION MEMBERSHIPS | <b>Learning the Universe</b><br>Simons Foundation collaboration developing next-generation cosmological simulations, machine learning methods, and advanced Bayesian inference for large-scale surveys<br><i>Contributions:</i> Training set generation and BORG team<br><b>CAMELS</b><br>Collaboration building large public suites of cosmological hydrodynamical and <i>N</i> -body simulations spanning broad cosmological and astrophysical parameter ranges to train and test machine learning methods<br><i>Contributions:</i> Leading projects on semi-analytic modelling and simulation-based inference<br><b>Aquila Consortium</b><br>International collaboration developing Bayesian field-level inference techniques to reconstruct initial conditions and study large-scale structure formation<br><i>Contributions:</i> Leading projects on data generation, benchmarking, large-scale flows and cosmological parameters | 2025 – present<br>2025 – present<br>2022 – present       |

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|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| TEACHING,<br>MENTORING &<br>SERVICE    | <b>Student Supervision</b>                                                                                                                                                                                                                                                                          |                               |
|                                        | <i>Noah Pierce</i> (BSc, Southampton; with H. Desmond & T. Yasin)<br>“Field-Level Inference of the Galaxy Luminosity Function”                                                                                                                                                                      | 2025 – present                |
|                                        | <i>Joshua Darne</i> (MPhys, Oxford; with T. Yasin & H. Desmond)<br>“Radial acceleration relation in the NewHorizon hydrodynamical simulation”                                                                                                                                                       | 2024                          |
|                                        | <i>Fedir Boreiko</i> (BSc, Manchester; with T. Yasin & H. Desmond)<br>“The correlation between light and dark matter across cosmic time” [arXiv:2601.07799]                                                                                                                                         | 2024                          |
|                                        | <i>Enoch Ko</i> (BSc, Warwick; with T. Yasin & H. Desmond)<br>“Dark matter and galaxy dynamics: enduring puzzles” [arXiv:2508.03569]                                                                                                                                                                | 2024                          |
|                                        | <i>Catherine Gallagher</i> (MPhys, Oxford; with T. Yasin & H. Desmond)<br>“The influence of cosmic environment on galaxy properties” [arXiv:2503.14732]                                                                                                                                             | 2023 – 2024                   |
|                                        | <i>James Harvey</i> (BSc, Oxford; with T. Yasin & H. Desmond)<br>“Machine learning the time of last major merger from spectroscopic data”                                                                                                                                                           | 2023 – 2024                   |
|                                        | <b>Teaching Experience</b>                                                                                                                                                                                                                                                                          |                               |
|                                        | Tutor, <i>MPhys C1 Astrophysics</i> , University of Oxford<br>Graduate tutorials in cosmology and stellar astrophysics                                                                                                                                                                              | 2023 – 2024                   |
|                                        | Mentor, <i>Lumiere Education</i><br>Supervision of senior high school students carrying out independent research projects                                                                                                                                                                           | 2023 – present                |
|                                        | Demonstrator, <i>3rd-Year Practical Course</i> , University of Oxford<br>Computational astrophysics practical sessions                                                                                                                                                                              | 2023                          |
|                                        | <b>Academic Service</b>                                                                                                                                                                                                                                                                             |                               |
|                                        | Referee for <i>A&amp;A</i> , <i>ApJ</i> , <i>MNRAS</i> , <i>OJA</i> , <i>PNAS</i> , and <i>PRD</i><br>Reviewed 20+ manuscripts for leading journals                                                                                                                                                 | 2022 – present                |
| SELECTED<br>AWARDS AND<br>SCHOLARSHIPS | Coordinator, Aquila Consortium Monthly Telecon<br>Organiser, “Simulator’s Coffee”, University of Oxford                                                                                                                                                                                             | 2023 – present<br>2023 – 2024 |
|                                        | Local organiser, Aquila Consortium Oxford Meeting                                                                                                                                                                                                                                                   | 2023                          |
|                                        | Lead organiser, “Middle of Scotland Science Festival”                                                                                                                                                                                                                                               | 2018                          |
|                                        | Hintze Prize Fellowship in Astrophysics (Cosmology), University of Oxford                                                                                                                                                                                                                           | 2026                          |
|                                        | Flatiron Research Fellowship, Flatiron Institute (declined)                                                                                                                                                                                                                                         | 2026                          |
|                                        | Junior Research Fellowship, Emmanuel College, Cambridge (declined)                                                                                                                                                                                                                                  | 2026                          |
|                                        | Snell Exhibition, Balliol College                                                                                                                                                                                                                                                                   | 2022 – 2026                   |
|                                        | STFC PhD Funding, Science and Technology Facilities Council                                                                                                                                                                                                                                         | 2022 – 2026                   |
|                                        | DAAD Study Scholarship, German Academic Exchange Service                                                                                                                                                                                                                                            | 2021 – 2022                   |
|                                        | Kerr Bursary, University of Glasgow                                                                                                                                                                                                                                                                 | 2020                          |
|                                        | Lang Scholarship, University of Glasgow                                                                                                                                                                                                                                                             | 2019                          |
|                                        | Undergraduate Summer Bursary, Royal Astronomical Society                                                                                                                                                                                                                                            | 2018                          |
|                                        | Dean’s List, Hong Kong University of Science and Technology                                                                                                                                                                                                                                         | 2018                          |
| SKILLS                                 | Astronomy 1 Prize, University of Glasgow                                                                                                                                                                                                                                                            | 2017                          |
|                                        | Matthew A. Muir Bursary, University of Glasgow                                                                                                                                                                                                                                                      | 2017                          |
|                                        | South East Asia Study Abroad Scholarship, University of Glasgow                                                                                                                                                                                                                                     | 2017 – 2018                   |
|                                        | <b>Programming:</b> Python, Julia, C/C++, Fortran; high-performance computing (MPI, GPU acceleration)                                                                                                                                                                                               |                               |
|                                        | <b>Statistics &amp; Machine Learning:</b> Bayesian inference (NumPyro/JAX, PyMC, Stan), simulation-based inference, density estimation; deep learning (MLPs, CNNs, GNNs, normalising flows) with JAX, TensorFlow, PyTorch; tree-based methods (random forests, gradient boosting) with scikit-learn |                               |
|                                        | <b>Astrophysics Software:</b> RAMSES, AREPO, Gadget, Rockstar, DisPerSe; <i>N</i> -body/hydrodynamical simulation analysis                                                                                                                                                                          |                               |
|                                        | <b>Languages:</b> English (fluent), Czech (native), Slovak (fluent), French (intermediate), German (beginner)                                                                                                                                                                                       |                               |

**First-Author – Submitted**

- [1] “Learning the Universe with cosmological rescaling of merger trees and semi-analytic galaxy formation models”; **R. Stiskalek**, L. A. Perez, S. Genel, R. S. Somerville, R. E. Angulo, S. Contreras [arXiv:2606.10024]
- [2] “Forward-modelling Milky Way Cepheids: selection effects and physical priors in the Gaia–HST calibration”; **R. Stiskalek**, A. Riess, H. Desmond, G. Lavaux, D. Scolnic [arXiv:2603.09880]
- [3] “ $S_8$  from Tully–Fisher, fundamental plane, and supernova distances agree with *Planck*”; **R. Stiskalek** [arXiv:2509.20235]

**First-Author – Accepted**

- [4] “Validating Digital Twins of the Local Universe with the Thermal Sunyaev-Zel’dovich Signal”; **R. Stiskalek**, H. Desmond [Open Journal of Astrophysics 9, arXiv:2601.15935]
- [5] “Revisiting the Great Attractor: The Local Group’s streamline trajectory, cosmic velocity and dynamical fate”; **R. Stiskalek**, H. Desmond, S. McAlpine, G. Lavaux, J. Jasche, M. J. Hudson [Open Journal of Astrophysics 9, arXiv:2601.08524]
- [6] “1.8 per cent measurement of  $H_0$  from Cepheids alone”; **R. Stiskalek**, H. Desmond, E. Tsaprazi, A. Heavens, G. Lavaux, S. McAlpine, J. Jasche [MNRAS 546:staf2260, arXiv:2509.09665]
- [7] “No evidence for local  $H_0$  anisotropy from Tully–Fisher or supernova distances”; **R. Stiskalek**, H. Desmond, G. Lavaux [MNRAS 546:staf2048, arXiv:2509.14997]
- [8] “The Velocity Field Olympics: Assessing velocity field reconstructions with direct distance tracers”; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz, G. Lavaux, M. J. Hudson, D. J. Bartlett, H. Courtois [MNRAS 545:staf1960, arXiv:2502.00121]
- [9] “Testing the local supervoid solution to the Hubble tension with direct distance tracers”; **R. Stiskalek**, H. Desmond, I. Banik [MNRAS 543:1556, arXiv:2506.10518]
- [10] “Symmetry in Hyper Suprime-Cam galaxy spin directions”; **R. Stiskalek**, H. Desmond [Res. Notes AAS 8 281, arXiv:2410.18884]
- [11] “Evaluating the variance of individual halo properties in constrained cosmological simulations”; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz [MNRAS 534:3120, arXiv:2310.20672]
- [12] “Probing general relativistic spin-orbit coupling with gravitational waves from hierarchical triple systems”; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [MNRAS 535:L1, arXiv:2307.01903]
- [13] “On the fundamentality of the radial acceleration relation for late-type galaxy dynamics”; **R. Stiskalek**, H. Desmond [MNRAS 525:6130, arXiv:2305.19978]
- [14] “Frequency- and polarization-dependent lensing of gravitational waves in strong gravitational fields”; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [Phys. Rev. D 109, 124045, arXiv:2209.06459]
- [15] “The scatter in the galaxy–halo connection: a machine learning analysis”; **R. Stiskalek**, D. J. Bartlett, H. Desmond, D. Anbajagane [MNRAS 514:4026, arXiv:2202.14006]
- [16] “The dependence of subhalo abundance matching on galaxy photometry and selection criteria”; **R. Stiskalek**, H. Desmond, T. Holvey, M. G. Jones [MNRAS 506:3205, arXiv:2101.02765]
- [17] “Are stellar-mass binary black hole mergers isotropically distributed?”; **R. Stiskalek**, J. Veitch, C. Messenger [MNRAS 501:970, arXiv:2003.02919]

**Co-Author – Submitted**

- [1] “Learning the Universe: Constrained simulations of the Coma galaxy cluster – I. Radial X-ray and Compton- $y$  signatures”; U. P. Steinwandel, S. McAlpine, **R. Stiskalek**,

Rüdiger Pakmor, V. Springel, E. Churazov, I. Khabibullin, J. Jasche, G. Lavaux, G. L. Bryan [arXiv:2606.10028]

- [2] “Introducing sapphire: Towards Hybrid Physics-Informed, Data-Driven Modeling of Galaxy Formation”; V. Pandya, . . . , **R. Stiskalek**, . . . [arXiv:2604.06318]
- [3] “Parameterizing Dark Energy at the density level: A two-parameter alternative to CPL”; G. Montefalcone, **R. Stiskalek** [arXiv:2603.25735]
- [4] “Testing cosmic anisotropy with cluster scaling relations”; T. Yasin, **R. Stiskalek**, H. Desmond, S. von Hausegger, P. G. Ferreira [arXiv:2602.06007]
- [5] “Constraining cosmological simulations with peculiar velocities: a forward-modelling approach”; A. Valade, N. Libeskind, D. Pomarède, **R. Stiskalek**, Y. Hoffman, S. Gottlöber, R. B. Tully [arXiv:2602.03699]
- [6] “Testing subhalo abundance matching with galaxy kinematics”; F. Boreiko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [arXiv:2601.07799]
- [7] “MEGATRON: Reproducing the Diversity of High-Redshift Galaxy Spectra with Cosmological Radiation Hydrodynamics Simulations”; H. Katz, . . . , **R. Stiskalek**, . . . [arXiv:2510.05201]
- [8] “MEGATRON: how the first stars create an iron metallicity plateau in the smallest dwarf galaxies”; M. Rey, . . . , **R. Stiskalek**, . . . [arXiv:2510.05232]
- [9] “MEGATRON: the impact of non-equilibrium effects and local radiation fields on the circumgalactic medium at cosmic noon”; C. Cadiou, . . . , **R. Stiskalek**, . . . [arXiv:2510.05667]

#### Co-Author – Accepted

- [10] “The subtle statistics of the distance ladder: On the distance prior and selection effects”; H. Desmond, **R. Stiskalek**, J. A. Najera, I. Banik [in press, arXiv:2511.03394]
- [11] “The galaxy-environment connection revealed by constrained simulations”; C. Gallagher, T. Yasin, **R. Stiskalek**, H. Desmond, M. J. Jarvis [MNRAS 546:stag108, arXiv:2503.14732]
- [12] “Renzo’s rule revisited: A statistical study of galaxies’ baryon–dark matter coupling”; E. Ko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [MNRAS 544:4288, arXiv:2508.03569]
- [13] “CosmoBench: A Multiscale, Multiview, Multitask Cosmology Benchmark for Geometric Deep Learning”; N. Huang, **R. Stiskalek**, J.-Y. Lee, A. E. Bayer, C. C. Margossian, C. K. Jespersen, L. A. Perez, L. K. Saul, F. Villaescusa-Navarro [NeurIPS 2025, arXiv:2507.03707]
- [14] “The Manticore Project I: a digital twin of our cosmic neighbourhood from Bayesian field-level analysis”; S. McAlpine, J. Jasche, M. Ata, G. Lavaux, **R. Stiskalek**, C. S. Frenk, A. Jenkins [MNRAS 540:716, arXiv:2505.10682]
- [15] “The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics”; Eleonora Di Valentino, . . . , **R. Stiskalek**, . . . [Physics of the Dark Universe, Volume 49, 2025, 101965, arXiv:2504.01669]
- [16] “Inferring the Ionizing Photon Contributions of High-Redshift Galaxies to Reionization with JWST NIRCам Photometry”; N. Choustikov, **R. Stiskalek**, A. Saxena, H. Katz, J. Devriendt, A. Slyz [MNRAS 537:2273, arXiv:2405.09720]

#### INVITED TALKS

*Supernova-free local ladders for  $H_0$*

GDR CoPhy Episode 4, Clermont-Ferrand, France

Cosmology seminar, UCL

*Bridging Machine Learning and Cosmology at Scale*

PV Reconstruction online meeting

*Is the Hubble tension easing?*

University of Oxford

2026

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| <i>Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling</i><br>CAMELS + Baryon Pasting Workshop, University of Osaka, Japan<br><i>Velocity Field Olympics</i><br>Leibniz Institute for Astrophysics Potsdam, Germany                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| <i>Forward modelling of peculiar velocities and large-scale structure</i><br>Institut d'Astrophysique de Paris<br><i>Pinning down the local Universe with digital twins</i><br>University of Cambridge<br>Yale University<br><i>Do SH0ES Cepheids prefer the Planck Hubble constant?</i><br>National Astronomy Meeting, Durham<br><i>Towards accurate models and tests of the local Universe</i><br>Columbia University<br>Princeton University<br>Center for Computational Astrophysics, Flatiron Institute<br><i>Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling</i><br>Center for Computational Astrophysics, Flatiron Institute<br><i>Velocity Field Olympics</i><br>Center for Computational Astrophysics, Flatiron Institute<br>Cosmic Flows 2025, Brisbane | <b>2025</b> |
| <i>Velocity Field Olympics</i><br>University of Portsmouth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>2024</b> |
| <i>Search for the optimal dark matter halo density profile</i><br>University of Oxford<br><i>Is the radial acceleration relation a fundamental correlation?</i><br>University of Oxford                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>2023</b> |
| <i>Frequency and polarization dependent propagation of gravitational waves</i><br>University of Glasgow<br>LMU Munich<br>Max Planck Institute for Gravitational Physics, Potsdam<br><i>The scatter in the galaxy–halo connection</i><br>Baryon Pasters Collaboration meeting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>2022</b> |
| <i>The scatter in the galaxy–halo connection</i><br>LMU Munich                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>2021</b> |
| <i>Reversible-jump MCMC in gravitational-wave astronomy</i><br>Max Planck Institute for Gravitational Physics, Hannover<br><i>Are binary-black hole mergers isotropically distributed?</i><br>LIGO Scientific Collaboration Data Analysis telecon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>2020</b> |
| <i>The relation between galaxies and dark matter halos</i><br>University of Oxford                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>2019</b> |